

3.4.1 Integral Test Worksheet

Determine whether each of the following series converges or diverges. You may use any theorem from 3.4.1 or earlier.

1. $\sum_{k=1}^{\infty} \frac{1}{2k-1}$

Let $f(x) = \frac{1}{2x-1}$. We note that $f(x)$ is continuous, positive, and decreasing on $[1, \infty)$, so the integral test applies.

$$\int_1^{\infty} \left(\frac{1}{2x-1} \right) dx = \lim_{t \rightarrow \infty} \left(\int_1^t \left(\frac{1}{2x-1} \right) dx \right) = \lim_{t \rightarrow \infty} \left(\left(\frac{1}{2} \ln |2x-1| \right) \Big|_1^t \right) = \lim_{t \rightarrow \infty} \left(\frac{1}{2} \ln |2t-1| - \frac{1}{2} \ln |1| \right) = \infty$$

Therefore, by the integral test, since the integral $\int_1^{\infty} \left(\frac{1}{2x-1} \right) dx$ is divergent the series $\sum_{k=1}^{\infty} \frac{1}{2k-1}$ is divergent. ■

2. $\sum_{k=1}^{\infty} \frac{3^k}{14k^2}$

Let $f(x) = \frac{3^x}{14x^2}$. By l'Hopital's rule we note that $\lim_{x \rightarrow \infty} \frac{3^x}{14x^2} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\ln(3)3^x}{28x} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\ln(3)^2 3^x}{28} = \infty$. Therefore

$\lim_{k \rightarrow \infty} \frac{3^k}{14k^2} = \infty$ and by the Test for Divergence the series $\sum_{k=1}^{\infty} \frac{3^k}{14k^2}$ diverges. ■

3. $\sum_{n=1}^{\infty} \frac{2\sqrt{n}}{n^3}$

Let $f(x) = \frac{2\sqrt{x}}{x^3} = \frac{2}{x^{5/2}}$. We note that $f(x)$ is continuous, positive, and decreasing on $[1, \infty)$, so the integral test applies.

$$\int_1^{\infty} \left(\frac{2}{x^{5/2}} \right) dx = \lim_{t \rightarrow \infty} \left(\int_1^t \left(\frac{2}{x^{5/2}} \right) dx \right) = \lim_{t \rightarrow \infty} \left(\left(\frac{-4}{3} x^{-3/2} \right) \Big|_1^t \right) = \lim_{t \rightarrow \infty} \left(\left(\frac{-4}{3x^{3/2}} \right) \Big|_1^t \right) = \lim_{t \rightarrow \infty} \left(\frac{-4}{3t^{3/2}} + \frac{4}{3} \right) = \frac{4}{3}$$

Therefore, by the integral test, since the integral $\int_1^{\infty} \left(\frac{2}{x^{5/2}} \right) dx$ is convergent the series $\sum_{n=1}^{\infty} \frac{2\sqrt{n}}{n^3}$ converges. ■

4.
$$\sum_{k=1}^{\infty} \frac{(-1)^k \cdot 2^{k+1} \cdot 3^k}{7^{2k}}$$

Note that
$$\sum_{k=1}^{\infty} \frac{(-1)^k \cdot 2^{k+1} \cdot 3^k}{7^{2k}} = \sum_{k=1}^{\infty} \frac{2 \cdot (-1)^k \cdot 2^k \cdot 3^k}{49^k} = \sum_{k=1}^{\infty} \frac{2 \cdot (-6)^k}{49^k} = \sum_{k=1}^{\infty} 2 \left(-\frac{6}{49} \right)^k = \sum_{k=1}^{\infty} -\frac{12}{49} \left(-\frac{6}{49} \right)^{k-1}.$$

So this is a geometric series with $r = -\frac{6}{49}$ and $a = -\frac{12}{49}$. Since $|r| < 1$ this series converges and

$$\sum_{k=1}^{\infty} -\frac{12}{49} \left(-\frac{6}{49} \right)^{k-1} = \frac{-12/49}{1 - (6/49)} = \frac{-12/49}{55/49} = -\frac{12}{55}.$$

■

5.
$$\sum_{n=1}^{\infty} \frac{3n+2}{n(n+1)}$$
 (Hint: Use partial fractions to rewrite the function associated with this series)

Let $f(x) = \frac{3x+2}{x(x+1)}$. We note that $f(x)$ is continuous and positive on $[1, \infty)$. Using partial fraction

decomposition we see that $f(x) = \frac{3x+2}{x(x+1)} = \frac{2}{x} + \frac{1}{x+1}$, which shows us that on $[1, \infty)$ $f(x)$ will

continue to decrease since it is the sum of two decreasing fractions. Thus the integral test applies.

$$\int_1^{\infty} \left(\frac{3x+2}{x(x+1)} \right) dx = \lim_{t \rightarrow \infty} \left(\int_1^t \left(\frac{2}{x} + \frac{1}{x+1} \right) dx \right) = \lim_{t \rightarrow \infty} \left((2 \ln |x| + \ln |x+1|) \Big|_1^t \right) = \lim_{t \rightarrow \infty} (2 \ln |t| + \ln |t+1| - 2 \ln |1| - \ln |2|) = \infty$$

Therefore, by the integral test, since the integral $\int_1^{\infty} \left(\frac{3x+2}{x(x+1)} \right) dx$ is divergent the series $\sum_{n=1}^{\infty} \frac{3n+2}{n(n+1)}$ is divergent.

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6.
$$\sum_{n=1}^{\infty} \frac{3n+2}{n+1}$$

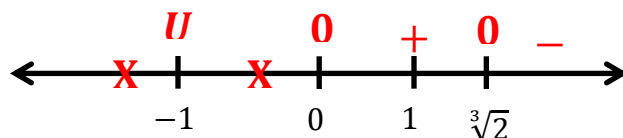
Note that $\lim_{n \rightarrow \infty} \frac{3n+2}{n+1} = \lim_{n \rightarrow \infty} \frac{3n+2}{n+1} \cdot \frac{1/n}{1/n} = \lim_{n \rightarrow \infty} \frac{3+2/n}{1+1/n} = \frac{3}{1} = 3$. Since this limit doesn't equal 0, then this series diverges by the Test for Divergence.

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7. $\sum_{n=1}^{\infty} \frac{n^2}{n^3+1}$

Let $f(x) = \frac{x^2}{x^3+1}$. We note that $f(x)$ is continuous and positive on $[1, \infty)$. We need to check to see if

$$f(x) \text{ is decreasing on } [a, \infty) \text{ so we will investigate } f'(x) = \frac{(x^3+1)2x - x^2 \cdot 3x^2}{(x^3+1)^2} = \frac{x(2-x^3)}{(x^3+1)^2}.$$



We can see that $f'(x) < 0$ for $x > \sqrt[3]{2}$. Therefore we can state that $f(x)$ is decreasing on $[2, \infty)$.

Therefore the integral test applies and we calculate the following:

$$\int_2^{\infty} \left(\frac{x^2}{x^3+1} \right) dx = \lim_{t \rightarrow \infty} \left(\int_2^t \left(\frac{x^2}{x^3+1} \right) dx \right) = \lim_{t \rightarrow \infty} \left(\frac{1}{3} \int_9^{t^3+1} \left(\frac{1}{u} \right) du \right) = \lim_{t \rightarrow \infty} \left(\frac{1}{3} (\ln |u|) \Big|_9^{t^3+1} \right) = \frac{1}{3} \lim_{t \rightarrow \infty} (\ln |t^3+1| - \ln |9|) = \infty$$

Here we used the substitution: $u = x^3 + 1$ which implies $du = 3x^2 dx \rightarrow \frac{1}{3} du = x^2 dx$.

Therefore, by the integral test, since $\int_2^{\infty} \left(\frac{x^2}{x^3+1} \right) dx$ diverges, the series $\sum_{n=2}^{\infty} \frac{n^2}{n^3+1}$ is divergent and therefore

clearly the series $\sum_{n=1}^{\infty} \frac{n^2}{n^3+1}$ diverges since it contains only one additional term.

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